

Year 3 Science Mission: Seeds of Science

National Science Week 2026 · 15–23 August



Our Theme

"Seeds of Science: Nurturing
knowledge for all"



Your Mission

Think like a scientist. Ask
questions. Discover!

Ready to think like a scientist? 



What Is National Science Week?

A National Celebration

Every August, Australia celebrates science! Schools, museums and communities come together to explore, question and discover.

This Year's Theme

"Seeds of Science: Nurturing knowledge for all" — just like a seed grows with the right care, ideas grow when we ask questions and share what we learn.

Why Do Scientists Ask Questions?

Questions lead to investigations. Investigations lead to evidence. Evidence helps us understand the world!

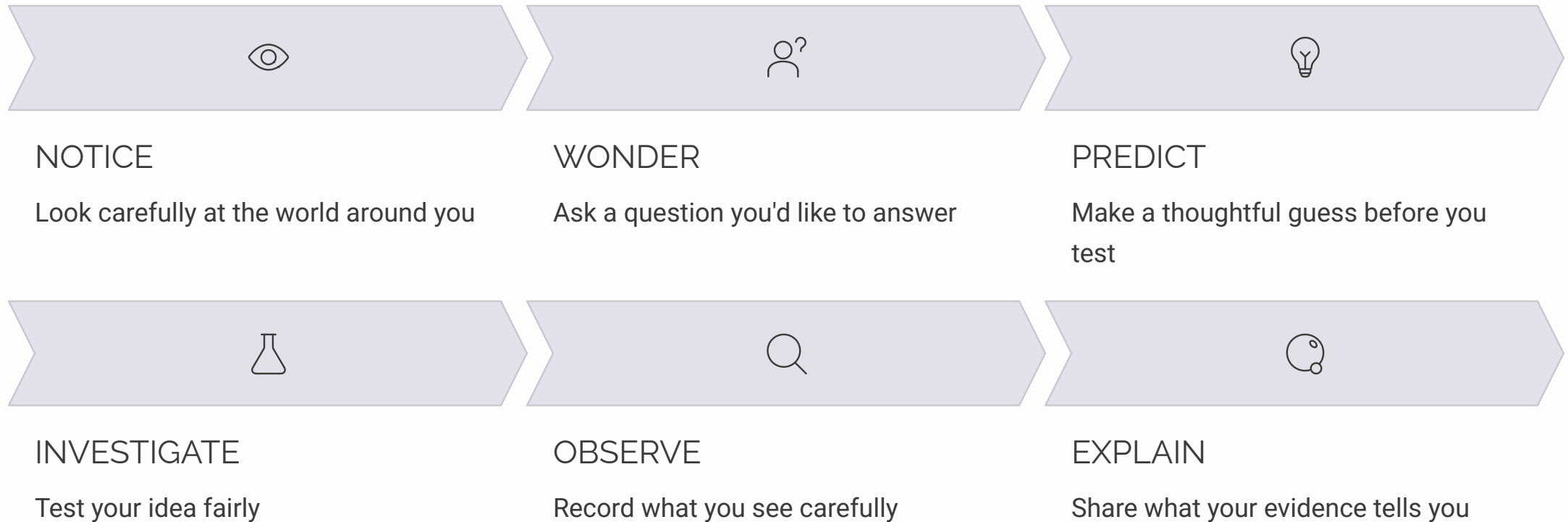
  **Did you know?** National Science Week reaches over **3 million Australians** every year — that's a LOT of curious minds!

 *"Seeds of Science is about inspiring the next generation to plant ideas, ask questions and grow innovative solutions."*

 **Your turn:** What would *you* like to investigate this Science Week? Turn to a partner and share your biggest science question!

Think Like a Scientist

Scientists don't just read answers — they **find** them! Here's the inquiry process we use in Year 3 Science:



 **Example in action:** You notice a seedling near the window is leaning towards the light. You wonder: *"Do plants always grow towards light?"* You predict: *"Yes, because plants need light to grow."* Then you plan a fair investigation to test it!

  **Quick predict:** If you planted two seeds — one in a sunny spot and one in a dark cupboard — what do you think would happen? *Why?*

Life Cycles: Growing & Changing

Every living thing goes through a **life cycle** — a series of stages from birth to death and back to new life. This connects perfectly to our theme: *Seeds of Science!*

A Plant's Life Cycle

01

Seed

A tiny seed contains everything needed to begin life

02

Germination

The seed absorbs water and a shoot breaks through the soil

03

Seedling

Roots grow down, stem and leaves grow up towards sunlight

04

Mature Plant

The plant grows flowers, which are pollinated to make new seeds

A Butterfly's Life Cycle

01

Egg

Laid on a leaf by the adult butterfly

02

Larva (Caterpillar)

Hatches and eats leaves to grow bigger

03

Pupa (Chrysalis)

The caterpillar changes inside a protective case

04

Adult Butterfly

Emerges, flies, and lays new eggs to begin the cycle again

 **Challenge:** Can you put these stages in order? *Adult* → *Egg* → *Caterpillar* → *Pupa* — write the correct order in your science journal!

Day, Night & Shadows

Why Do We Have Day and Night?

Earth spins on its axis — one full rotation takes **24 hours**. The side facing the Sun experiences **day**. The side facing away experiences **night**.

What Causes Shadows?

A shadow forms when an object **blocks light**. The position of the light source changes where and how long the shadow is:

 Sun is **LOW** (morning or afternoon)

Shadows are **long** because light hits objects at a sharp angle

 Sun is **HIGH** (midday)

Shadows are **short** because light hits objects almost straight down



Prediction time!

When the Sun is *higher* in the sky, what happens to a shadow?

- A) Gets longer
- B) Gets shorter
- C) Stays the same

Think about the angle of the light — what's your prediction? Why?

Heat & Temperature

Heat is a type of ENERGY

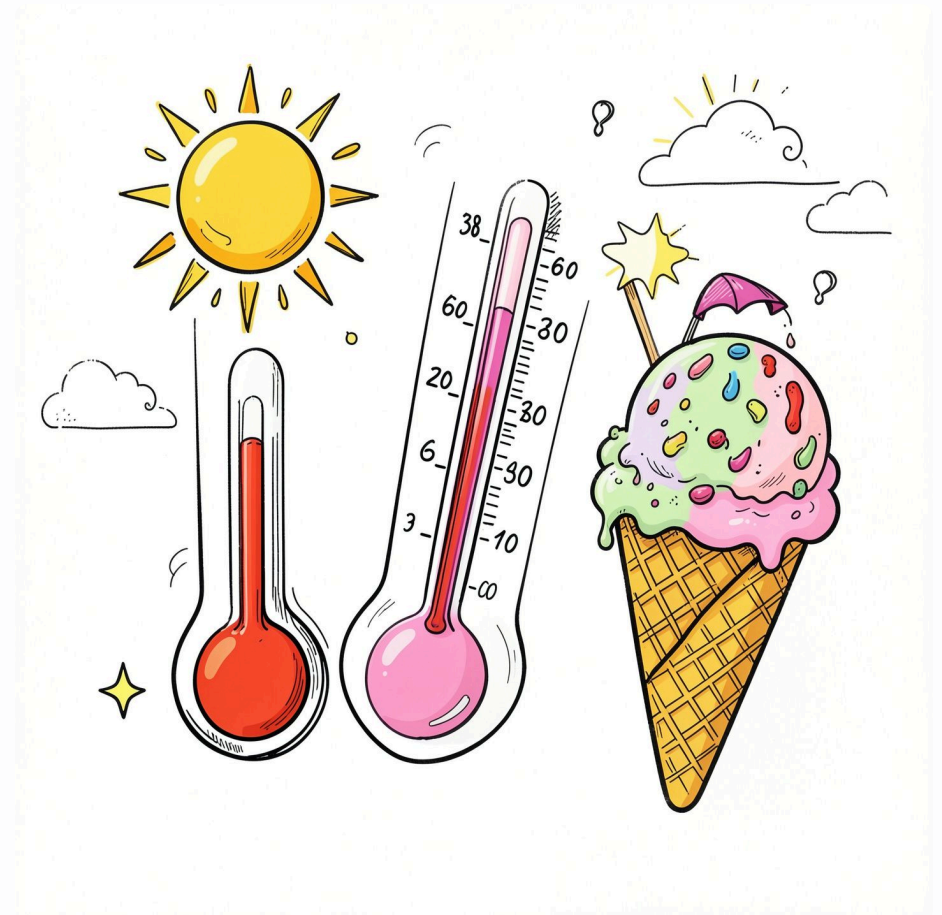
Heat moves from **warmer** objects to **cooler** ones. For example, a warm mug warms your cold hands — energy is flowing!

Temperature is a MEASUREMENT

Temperature tells us **how hot or cold** something is. We measure it in degrees Celsius ($^{\circ}\text{C}$) using a thermometer.

Everyday examples:

- The Sun warms the footpath — the footpath gets hot (gains heat energy)
- An ice block melts in your hand — heat flows from your hand into the ice
- A thermometer in a warm room shows a higher temperature than one outside on a cold day



📋 🎯 Comparison challenge!

Which would you expect to feel warmer — a metal spoon left in the sun, or a wooden spoon left in the sun?

Explain your reasoning using what you know about heat!

(Hint: both are in the same sunlight — think about the material!)

Living or Non-Living?

Scientists use specific characteristics to classify whether something is living or non-living.

✓ Living things can:

→ Grow & change

They develop over time through a life cycle

→ Respond to their environment

They react to light, temperature, touch and more

→ Reproduce

They make more of their own kind (seeds, eggs, babies)

→ Use energy

They need food, water or sunlight to survive

🎯 Classification Challenge!

Sort these into **Living** or **Non-Living**. Some might surprise you!

 Plant

Living – grows, reproduces, uses energy from sunlight

 Dog

Living – breathes, grows, responds, reproduces

 Rock

Non-living – does not grow, respond or reproduce

 Mushroom

Living – a fungus! It grows and reproduces using spores

 Toy

Non-living – made by humans, does not grow or respond

 Tree

Living – grows, produces seeds, responds to sunlight

Note: A seed is alive even when dormant, and some tricky cases (like fire or crystals) can look like they are living – scientists look for ALL the characteristics, not just one!

Mini Science Investigation: Shadow Lengths

? Question

How does shadow length change during the school day?

🤔 Prediction

I predict that shadows will be **longest** in the ____ and **shortest** at _____. This is because _____.

🧰 Materials

A long stick or ruler · Chalk or tape measure · Recording table · Clock or timer

📋 Procedure

1. Place a stick upright in a sunny spot (same spot all day)
2. At **3 different times** (e.g. 9am, 12pm, 3pm), measure the shadow length in centimetres
3. Record your results in your table
4. Draw the shadow position each time

📊 Recording Table

Time	Shadow Length (cm)	Shadow Direction
9:00 am		
12:00 pm		
3:00 pm		

⚖️ **Fair-test reminder:** Keep the *same stick, same spot, same measuring method* each time!



Safety reminder: NEVER look directly at the Sun. Use a stick or a ruler to measure shadows only.

Science Week Challenge Questions

Use your scientific thinking – don't just recall facts, **explain your reasoning!**

- | | |
|---|---|
| 1 | <p>Life Cycles</p> <p>A caterpillar and a butterfly look completely different, but they are the same animal at different stages. Why is it important for scientists to understand life cycles when studying living things?</p> |
| 2 | <p>Day & Night</p> <p>If Earth stopped spinning, one side would always face the Sun. What do you think would happen to plants and animals on the dark side? Use what you know about living things to explain.</p> |
| 3 | <p>Shadows</p> <p>Nadia measures her shadow at 8am and it is 3 metres long. At 12pm it is only 0.5 metres long. What caused this change? What would happen to the shadow at 4pm?</p> |
| 4 | <p>Heat & Temperature</p> <p>Two cups of water are left in the sun – one is covered with foil and one is left open. After 30 minutes, which would you predict is warmer? What evidence would you collect to test this fairly?</p> |
| 5 | <p>Living or Non-Living?</p> <p>A flame grows, moves and uses energy – but is it alive? Use the characteristics of living things to explain your answer clearly.</p> |

Teacher note: Suggested answers in speaker notes below ↓

Your Science Mission

Be curious. Ask questions. Test ideas. Look closely. Explain what you discover.



NOTICE it

Find one thing around you and look closely



Make a PREDICTION

What do you think will happen? Why do you think that?



EXPLAIN your discovery

Share what the evidence tells you — that's real science!



WONDER about it

Ask a science question — what do you want to know?



INVESTIGATE

Test your idea — keep it fair and record what you find

 **Every big scientific discovery starts with a question.**

Choose one thing around you. Notice it. Wonder about it. Make a prediction. Then investigate.

Happy Science Week from Aussie Primary Academy!  

